

GENERAL SPECIFICATION G5.36

Distribution Panels

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G5.36 DISTRIBUTION PANELS

G5.36.1 General

(a) Introduction

- (i) Perform all work; design, supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, as listed in General Specification G5.2- Standards, Codes and Regulations.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

1) British Standards Institute (BSI)

Standard	Subject
BS 5252	Framework for Colour Co-Ordination for Building Purposes.
BS EN 61008	Specification for Residual Current Operated Circuit-Breakers Without Integral Overcurrent Protection for Household and Similar Uses (RCCBs).

2) International Electro-technical Commission (IEC)

Standard	Subject
IEC 60044	Instrument transformers.
IEC 60051	Direct Acting Indicating Analogue Electrical Measuring Instruments and Their Accessories.
IEC 60228	Conductors of insulated cables.
IEC 60238	Edison screw lamp holders.
IEC 60255	Measuring Relays and Protection Equipment.
IEC 60364	Low Voltage Electrical Installations.
IEC 60439	Low Voltage Switchgear and Control gear Assemblies.
IEC 60502	Power cables with extruded insulation and their accessories for rated voltages from 1 kV (Um = 1,2 kV) up to 30 kV (Um = 36 kV)
IEC 60529	Degrees of Protection Provided by Enclosures.

Standard	Subject
IEC 60715	Dimensions of Low-Voltage Switchgear and Control gear. Standardised Mounting on Rails for Mechanical Support of Electrical Devices in Switchgear and Control gear Installations.
IEC 60898	Electrical Accessories - Circuit-Breakers for Overcurrent Protection for Household and Similar Installations.
IEC 60947	Low-Voltage Switchgear and Control gear.
IEC 61008	Residual Current Operated Circuit-Breakers Without Integral Overcurrent Protection for Household and Similar Uses (RCCBs).
IEC 61032	Protection of Persons and Equipment by Enclosure – Probes for Verification.
IEC 61439	Low-voltage switchgear and control gear assemblies
IEC 61558	Safety of power transformers, power supplies, reactors and similar products.
IEC/TR 61641	Enclosed low-voltage switchgear and control gear assemblies - Guide for testing under conditions of arcing due to internal fault.
IEC 61643	Surge Protective Devices Connected to Low-Voltage Power Distribution Systems.

3) Spring Singapore

Standard	Subject
SS 97	Residual Current Operated Circuit Breakers (RCCBs).
SS 480	Residual Current Operated Circuit-Breaker With Integral Overcurrent Protection for Household and Similar Uses (RCBOs).
CP 5	Code of Practice for Electrical Installations.
SS 551	Code of Practice for Earthing.

- (ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..
- (c) Submittals
 - (i) Shop Drawings, and Technical Data: As a minimum, submit the following items:
 - 1) Descriptive product information;
 - 2) Itemised Bill of Material;
 - 3) Dimensional drawings;
 - 4) Safe switching and operating procedures;
 - 5) Anchoring instructions and details;
 - 6) Circuit Breakers: Copies of time-current characteristics;
 - 7) Earth Fault Protection: Relay time-current characteristics;
 - 8) Bus data;
 - 9) Cable entrance locations;
 - 10) Heat dissipation and temperature rise calculations;
 - 11) Manufacturer's installation instructions;
 - 12) Type test certificate for short-circuit test for overall assembly;
 - 13) Type test certificate for bulbar pack; and
 - 14) Operation and Maintenance Manual.
- (d) Definitions
 - (i) Distribution Board (DB): A distribution board having a main circuit breaker not exceeding 300 amperes and equipped with moulded case circuit breakers (MCCB) and miniature circuit breakers (MCB). Provide residual current operated circuit breakers (RCBO) in series ahead of the final circuits where shown in drawings and / or panel schedules. Refer to typical DB schematic drawing for details.

G5.36.2 Products

- (a) Manufacturers
 - (i) Qualification
 - 1) Normally engaged in the manufacturer of distribution and sub-distribution electrical panels;
 - 2) Minimum of 10 years' experience in the design and manufacturer of distribution panels;
 - 3) Certified under ISO 9001 for manufacturer of electrical apparatus;
 - 4) Manufacturers as listed in latest edition of the Spring Singapore "Directory of SS Certified Products."; and
 - 5) Fully Type tested design in accordance with IEC 61439 standards.

- (b) General
 - (i) Service: 400/230 volts, three-phase, four-wire earthed wye.
 - (ii) Designed and assembled in accordance with IEC 60439-1 and IEC 61439.
 - (iii) Operating Conditions
 - 1) Ambient Temperature: Maximum 40 degrees C;
 - 2) Elevation: <1000 m above Mean Sea Level; and
 - 3) Equipment shall be fully rated without de-rating for operating conditions.
 - (iv) Provide CLUSTERED LED 3-phase indicating lamps (white, white, white) connected before and after the main circuit breaker for INCOMING and OUTGOING indicating lights respectively.
 - (v) All incoming and critical outgoing circuit breaker status shall be monitored in SCADA / PMS / EMS regardless of whether they are shown in drawings or I/O list. Provide necessary FNI / RIO along with cabling works as necessary.
 - (vi) Only major loads are shown in SLD and / or DB Schedule. Contractor shall provide suitably rated RCBO in outgoing circuits for all other miscellaneous loads including but not limited to equipment package systems, control panels, associated lighting and small power, etc. required for a fully functional system. All costs associated with component, installation, cabling works, etc. shall be included in the contract value.
 - (vii) 20% spare feeders shall be provided as minimum for each rating based on final loads.
 - (viii) All DB's including LDB's shall be from a single manufacturer.
- (c) Enclosure
 - (i) Material: Hot-dip galvanised sheet steel, minimum thickness of 2 mm, with reinforced steel frame.
 - (ii) Finish: Baked enamel applied over a rust-inhibiting, phosphate base coating:
 - 1) Colour:
 - Exterior: Grey, BS 5252, Colour Code 00.A.05, finish gloss, or RAL 7035 with matt finish;
 - Interior: Manufacturer's standard; and
 - Unpainted Parts: Plated for corrosion resistance.
 - (iii) Interior
 - 1) Factory assembled; complete with circuit breakers;
 - 2) Capable of circuit breaker replacement without disturbing adjacent circuit breakers;
 - 3) Supplied with separate earth and neutral connectors;
 - 4) Dead front with the door open; and
 - 5) It shall be possible to change branch circuit load connections without exposing personnel to any line-side bus or terminals.

- (iv) Door: Hinged door with concealed hinges.
- (v) Degree of protection: IP 55 (minimum) unless otherwise shown.
- (vi) Oversized enclosures shall be provided where required to allow entrance of a minimum of at least one cable for each circuit breaker, through a cable gland fitting, through the bottom of the enclosure.
- (d) Hazardous Areas
 - (i) Hazardous Areas are areas where explosive gas/air/dust/fibre mixtures are likely to exist. These are defined in the Hazardous Area Classification and in Hazardous Area Zone Drawings. The Contractor shall:
 - (ii) Provide materials and equipment acceptable to the regulatory authority having jurisdiction for the level of hazard indicated, and in accordance with IEC 60079.
 - (iii) All hazardous area classified apparatus shall have either ATEX or IECEx certification
 - (iv) All hazardous area classified equipment / devices shall be certified for use the zone and conditions stipulated in the Hazardous Area Drawings and HAC.
 - (v) All equipment for hazardous areas shall be supplied with either an EC Declaration of Conformity to ATEX 95 Directive, or an IECEx certificate of conformity. One certificate for each model used shall be sufficient. Original certificates shall be arranged in a dedicated file. Copies shall be included in the Operation & Maintenance Manuals.

“Ex” warning signs shall be installed at hazardous area
- (e) Buswork And Busbar Pack
 - (i) Busbar pack when specified shall be a plug-in socket system where protection devices are snapped in and modular construction.
 - (ii) Quick and easy exchangeability of the devices without need to shutdown the DB.
 - (iii) Easy reassignment of loads between phases.
 - (iv) Flexibility of mixed-pole arrangement of all devices next to each other.
 - (v) Busbar Material: Tinned Copper, full size throughout length.
 - (vi) Provide auxiliary busbars for alarm and signal contacts.
 - (vii) Provide for connection of future busbar packs. Machine, drill and tap as required for current and future positions.
 - (viii) Ultimate Current Withstand Capacity:
 - 1) Short-circuit withstand rating of distribution boards shall be 65,000 amperes rms symmetrical unless otherwise indicated; and
 - 2) Short circuit withstand capacity shall be coordinated with all upstream and downstream breakers.
 - (ix) Neutral: Rated 100%, Tinned Copper, insulated from enclosure, rated same as phase bus bars with at least one terminal screw for each circuit breaker pole.

- (x) Earth: Rated 50%, Tinned Copper, bonded to enclosure, with at least one terminal screw for each circuit breaker pole or space.
- (xi) All upstream and downstream circuit breakers shall be Type 2 coordinated for protection.
- (f) Circuit Breakers
 - (i) Moulded Case Circuit Breakers (MCCB)
 - 1) Plug-in type, Moulded case, thermal-magnetic, quick make, quick break, trip-free, of the indicating type showing OFF/ON and tripped position of the operating handle;
 - 2) Pad-lockable in OFF position;
 - 3) In compliance with IEC 60947-1, IEC 60947-2;
 - 4) Ultimate Breaking Capacity (Icu): 65 kA (minimum) at 415 volts, AC;
 - 5) Ics not less than 100 percent of Icu;
 - 6) Front face of all circuit breakers shall be flush with each other;
 - 7) Multi-pole circuit breakers designed to automatically open all poles when an overload occurs on one pole;
 - 8) Two- or three-pole circuit breakers shall fit in the same space as two or three single-pole circuit breakers without requiring alteration of the distribution board or any wiring connections;
 - 9) Main incoming and outgoing circuit breakers in distribution panels shall be equipped with an earth fault relay operated shunt trip device;
 - 10) All incoming and outgoing circuit breakers shall be 4 pole unless specified otherwise; and
 - 11) Auxiliary contacts for monitoring of status at the SCADA / PMS / EMS.
 - (ii) Miniature Circuit Breakers (MCB)
 - 1) Plug-in type, Moulded case, thermal-magnetic, quick make, quick break, trip-free, of the indicating type showing OFF/ON and tripped position of the operating handle;
 - 2) In compliance with IEC 60947-1, IEC 60947-2;
 - 3) Ultimate Breaking Capacity (Icu): 50 kA (minimum) at 415 volts, AC;
 - 4) Ics not less than 75 percent of Icu;
 - 5) Front face of all circuit breakers shall be flush with each other;
 - 6) Multi-pole circuit breakers designed to automatically open all poles when an overload occurs on one pole;
 - 7) Two-, three- or four-pole circuit breakers shall fit in the same space as two, three or four single-pole circuit breakers without requiring alteration of the distribution board or any wiring connections; and
 - 8) Auxiliary contacts for monitoring of status at the SCADA.

- (iii) Residual Current Circuit Breaker (RCCB)
 - 1) Plug-in type, Moulded case circuit breaker with electromagnetic release in case of earth fault;
 - 2) Trip Rating: 30 mA, 100 mA or 300 mA or as required;
 - 3) Positive ON/OFF indication;
 - 4) TRIP indication on front face by red flag;
 - 5) Voltage Rating (Minimum): 240 volts for two-pole breakers and 415 volts for four-pole breakers;
 - 6) Tropicalised;
 - 7) Compliant with SS 97; and
 - 8) Short-circuit rating selected in co-ordination with short-circuit protective devices as indicated in Clause 5.4 of SS 97: Part I.
- (iv) Residual Current Operated Circuit Breaker (RCBO)
 - 1) Plug-in type, Moulded case circuit breaker with both thermal overload protection and electromagnetic release in case of earth fault;
 - 2) Trip Rating on Earth Fault: 30 mA, 100 mA or 300 mA unless otherwise shown;
 - 3) Positive ON/OFF indication;
 - 4) TRIP indication on front face by red flag;
 - 5) Voltage Rating (Minimum): 240 volts for two-pole breakers and 415 volts for four-pole breakers;
 - 6) Tropicalised;
 - 7) Compliant with SS 480, and IEC 61009-1;
 - 8) Short-Circuit Rating: 50 kA minimum; and
 - 9) Instantaneous Tripping Characteristics: Type B or as required by the device rating with range of 3 to and including 5 times nominal current rating of the device.
- (g) Surge Protective Device (SPD)
 - (i) SPDs shall comply with IEC 61643;
 - (ii) SPD Type-1 shall be provided for panels with main incoming in to the building or outdoor;
 - (iii) SPD Type-2 shall be provided for all other indoor panels;
 - (iv) All SPD shall be plug-in type;
 - (v) SPD shall be protected by MCB as recommended by SPD manufacturer; and
 - (vi) Coordination between upstream and downstream SPD shall be carried out by the contractor's LEW.

- (h) Energy Meter (Type EM-3) for incomers
 - (i) Microprocessor based device with easy to read LCD displays;
 - (ii) Direct voltage input up to 600 volts AC;
 - (iii) Current input via current transformer with 5 amperes secondary. All CT's shall have appropriate CT ratios with accuracy Class 0.5;
 - (iv) Programmable current and potential transformer ratios;
 - (v) Programmable limits to activate up to four alarms;
 - (vi) Additional 16 I/O capability (to monitor incoming breaker status, other critical outgoing breaker status, SPD status, etc.) with facility to extend;
 - (vii) Selectable Voltage Measurements: Line-to-line or line-to-neutral and wye or delta;
 - (viii) Simultaneous Display and Data Measurement
 - 1) Volts, three-phase;
 - 2) Amperes, three-phase;
 - 3) Kilowatts, Reactive power, and KVA;
 - 4) Kilowatt-hours (kWh), KVAh and KVARh;
 - 5) Power factor;
 - 6) Frequency; and
 - 7) Earth leakage mA.
 - (ix) Communications System Interface: Each energy metre shall be capable of communicating to a host computer system over an Ethernet network:
 - 1) Media Connection: Fibre optic;
 - 2) Protocol: Master-slave as defined by MODBUS RTU over RS-485;
 - 3) Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational;
 - 4) Data Registers: As a minimum, provide the following information as data which can be queried by the ModBus host:
 - All items listed for display;
 - All alarms;
 - Internal faults; and
 - Communication loss.
 - (x) Provide front panel accessible RS-232 port connection for local interrogation and programming using a portable PC (by others) and a Windows-based program specific to this unit. Program to enable full access to unit's data and provide, as a minimum, the following features:
 - 1) View actual values;
 - 2) View unit status;

- 3) View/edit settings (online and offline);
 - 4) View event record (for troubleshooting);
 - 5) Help screens and step-by-step tutorial; and
 - 6) Provision for installation of firmware upgrade(s) to unit.
- (xi) Degree of Protection: IP 51;
- (i) Equipment Identification
- (i) Master Nameplate
 - 1) Deep-etched aluminium with manufacturer's name and model number; and
 - 2) Riveted to enclosure.
 - (ii) Nameplate
 - 1) Engraved, acrylic or trefolite with distribution board name and power source;
 - 2) White with black block type characters;
 - 3) Character Height: 8 mm;
 - 4) Size: As required for required inscription;
 - 5) Inscriptions: As shown on single-line diagram or distribution board schedule;
 - 6) Blank plates for future spaces; and
 - 7) Attachment Screws: Stainless steel panhead.
- (j) Testing
- (i) General
 - 1) Test all Distribution Panels, including all components in accordance with IEC 61439; and
 - 2) Testing to be in accordance with established ISO 9001 procedures.
 - (ii) Factory Inspections and Testing
 - 1) Inspect for required construction, electrical connection, and intended function. All Source Quality Control and Factory Acceptance Tests (FAT) shall be witnessed by S.O..
 - 2) Notify S.O. at least 2 weeks prior to any testing to allow S.O. to witness tests.
 - 3) Completely assemble Distribution Panel at the factory, and perform following tests at manufacturer's shop in accordance with IEC 61439:
 - Carry out routine test as part of FAT (Factory Acceptance Test) in accordance with IEC Routine Tests on all Distribution Panels;
 - Conformance tests to demonstrate conformance of the equipment to applicable standards;

- Production tests to check the quality and uniformity of the workmanship and materials used in the manufacturer of the equipment; and
 - Verification of compliance of RCCBs and RCBOs with the requirements of electromagnetic compatibility (EMC) in accordance with EMC product family Standard for RCDs: IEC 1008-1.
- 4) Submit a follow-up testing program for RCCBs and RCBOs installed at Works in order to guarantee the keeping of the quality level of products.

G5.36.3 Execution

- (a) Installation
- (i) Install equipment in accordance with manufacturer's instructions and recommendations.
 - (ii) Install top of enclosure 1,800 mm above floor unless otherwise indicated.
 - (iii) Install securely, plumb, in-line and square with walls.
 - (iv) Seal all unused openings in enclosure.
 - (v) Distribution boards shall be fully wired and wiring shall be neatly installed and taped.
 - (vi) Provide phase identification colours at the front of all distribution boards.
 - (vii) Provide large, permanent, identifying numbers affixed to each circuit breaker.
 - (viii) Provide printed circuit directory for each distribution panel identifying the load served by each circuit breaker.
 - (ix) Prepare and provide a complete single-line diagram for each distribution panel and consumer unit.
 - (x) Miniature circuit breakers shall only be installed where protected by an RCCB or RCBO.
 - (xi) All cables entering a distribution boards or consumer units shall enter the bottom of the panel.
- (b) Field Testing
- (i) In accordance with General Specification G5.51- Electrical Testing.
- (c) Manufacturer's Services
- (i) Refer to General Specification G2.25 – Manufacturers' Services, and G2.26- Plant Testing and Commissioning.

END OF SECTION